

Question: Based on the given data, determine which material is most favorable for the stability of perovskite materials. The stability can be predicted by element properties such as density and first ionization potential.

Ground-Truth: B

(a) Traditional RL



To determine which perovskite material is most favorable for stability, we....
Therefore, the best choice is C. ❌



Reward=0



To determine which perovskite material is most favorable for stability, we....
Therefore, the best choice is B. ✅



Reward=1

(a) SocraticPO



To determine which perovskite material is most favorable for stability, we....
Therefore, the best choice is C. ❌



Reward=0

Your reasoning contains a critical contradiction in Step 2...
Next Step: (1) Re-check the density values for each material. (2) Re-evaluate the trend: ...



I apologize for the earlier error. Let's carefully re-examine....
Therefore, the most favorable for stability is B. ✅



Decayed Reward=0.7